**NCEA Level 2 Mathematics**

91261 · Apply algebraic methods in solving problems (Algebra)

*Excellence-level practice questions*

# PRACTICE ANSWER SCHEDULE

## About this answer schedule

- This schedule gives the **final answers only** to the 91261 practice excellence questions.
- Solutions are **NOT** provided in this answer schedule to encourage students **NOT** to just copy off the solution and think they learnt something.
- It is **not** an official NZQA assessment schedule, and no marks or grades are given. However, solving these levels of questions in the exam should award you an excellence mark.

## How to use it

- Only check an answer once you've made a proper attempt at the question.
- If your answer doesn't match, go back and find your mistake before looking at anything else.
- For "show that" questions, the result is given, so it's your working that counts. If you have proved a statement, then your statement counts as the answer, and counts as a solution.

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## Contact


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- If you spot an error or typo, please let me know so I can fix it.

Good luck!

# 91261 Practice Excellence Questions

Answer Schedule

*Solutions are NOT provided in this answer schedule to encourage students NOT to just copy off the solution and think they learnt something.*

| Q | Answer |
|---|---|
| 1 | $x = 2$ or $x = -1$ |
| 2 | $x = -1,\ 1,\ 2,\ 4$ |
| 3 | $x = 6$ |
| 4 | $\log_{ab} x = \frac{12}{7}$ |
| 5 | $x = \frac{\log 30}{\log 1.5}$ $(\approx 8.39)$ |
| 6 | After approximately 10.0 hours $(A \approx 1012,\ r \approx 1.581,\ t \approx 10.03)$ |
| 7 | After 21 complete years |
| 8 | Approximately 13.5 m |
| 9 | $\Delta = k^2 + 2k + 9 = (k+1)^2 + 8 > 0$ for all $k$, so two distinct real roots |
| 10 | $\Delta = (a-b)^2 + 4k^2 \geq 0$, so always real solutions; exactly one solution when $a = b$ and $k = 0$ |
| 11 | $m = 7$, touching at $(2, 13)$; $m = -1$, touching at $(-2, 1)$ |
| 12 | $3 < m < 4$ |
| 13 | $b = 7$ or $b = -7$ |
| 14 | Shown: roots $2r$ and $3r$ give $5r = -\frac{b}{a}$ and $6r^2 = \frac{c}{a}$, so $6b^2 = 25ac$ |
| 15 | $k = 0$ only |
| 16 | $k = 1$; not valid for $x = 4$ or $x = -4$ |
| 17 | 8 friends |
| 18 | $10\ \text{km h}^{-1}$ |
| 19 | 15 m |
| 20 | $x \approx 1.42$ |
| 21 | $x = \frac{3}{2},\ y = \frac{9}{4}$ |
| 22 | $x = 8,\ y = 4$ |
| 23 | $x < -\sqrt{6}$ or $x > \sqrt{6}$ |
| 24 | $x = 8$ or $x = 27$ |
| 25 | $x = 0$ or $x = 1$ |
| 26 | $x = 2$ or $x = 4$ |
| 27 | Shown: $n(n+1)(n+2)(n+3) + 1 = \left(n^2 + 3n + 1\right)^2$ |

| Q | Answer |
|---|---|
| 28 | (i) 23 (ii) 110 |
| 29 | $x = \dfrac{y+3}{y-2}$; $y$ can never equal 2 |
| 30 | $k = 1$ or $k = -1$ |
| 31 | $k = 0$ (solutions $x = -1,\ 0,\ 1$) |
| 32 | $10^{1.4} \approx 25.1$ times larger |
| 33 | Approximately 14 200 years ago ($\approx$ 14 176 years) |
| 34 | 5 cm and 12 cm |
| 35 | $\dfrac{L}{W} = \dfrac{1+\sqrt{5}}{2}$ |
| 36 | $\$10 \le p \le \$20$ |
| 37 | Ella is not correct: reciprocal roots need $2k = 1$, so $k = \frac{1}{2}$, but then $\Delta = -\frac{15}{4} < 0$ (no real roots) |
| 38 | $x = 3$ or $x = -1$ |
| 39 | $x = 100$ or $x = 0.1$ |
| 40 | $k = 1$ or $k = 2$ |